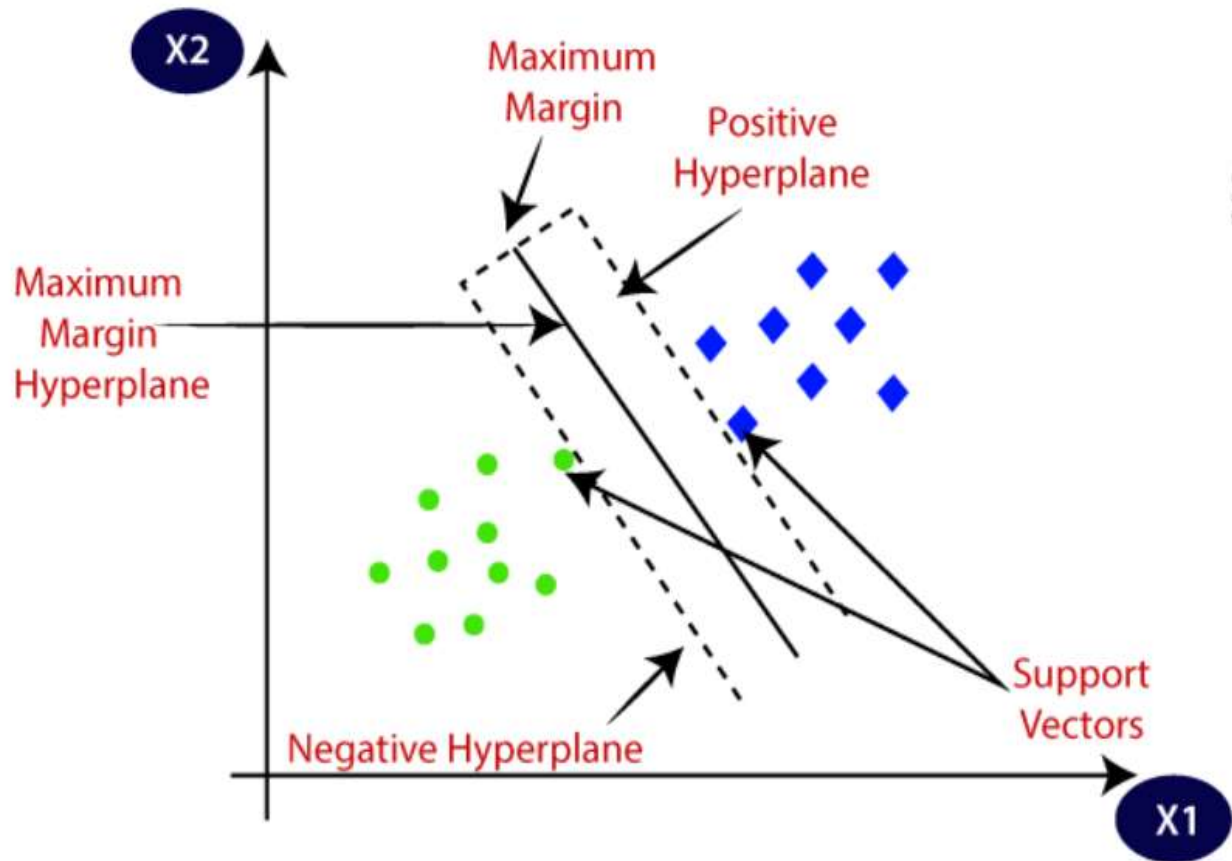


Support Vector Machine

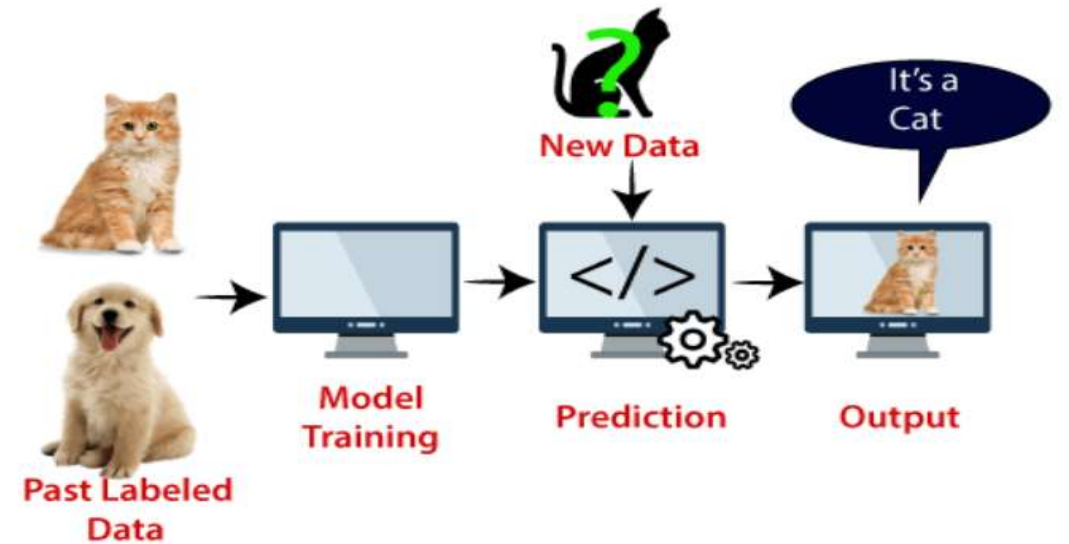
Support vector machine

- Support Vector Machine or SVM is one of the most popular **Supervised Learning algorithms**, which is used for Classification problems in Machine Learning.
- The goal of the SVM algorithm is to **create the best line or decision boundary that can segregate n-dimensional space into classes** so that we can easily put the new data point in the correct category in the future. **This best decision boundary is called a hyperplane.**
- SVM chooses the extreme points/vectors that help in creating the hyperplane. These extreme cases are called as **support vectors**, and hence algorithm is termed as **Support Vector Machine.**



- Boundary line situated at positive region is known as **Positive Hyperplane**
- The boundary line situated at negative region is known as **Negative Hyperplane**.

- Suppose we see a strange cat that also has some features of dogs, so if we want a model that can accurately identify whether it is a cat or dog, so such a model can be created by using the SVM algorithm.
- We will first train our model with lots of images of cats and dogs so that it can learn about different features of cats and dogs, and then we test it with this strange creature. So as support vector creates a decision boundary between these two data (cat and dog) and choose extreme cases (support vectors), it will see the extreme case of cat and dog. On the basis of the support vectors, it will classify it as a cat.



Types of SVM

SVM can be of two types:

- **Linear SVM:** Linear SVM is used for linearly separable data, which means if a dataset can be classified into two classes by using a single straight line, then such data is termed as linearly separable data, and classifier is used called as Linear SVM classifier.
- **Non-linear SVM:** Non-Linear SVM is used for non-linearly separated data, which means if a dataset cannot be classified by using a straight line, then such data is termed as non-linear data and classifier used is called as Non-linear SVM classifier.

Hyperplane and Support Vectors in SVM

Hyperplane: There can be multiple lines/decision boundaries to segregate the classes in n -dimensional space, but we need to find out the **best decision boundary** that helps to classify the data points. This best boundary is known as the hyperplane of SVM.

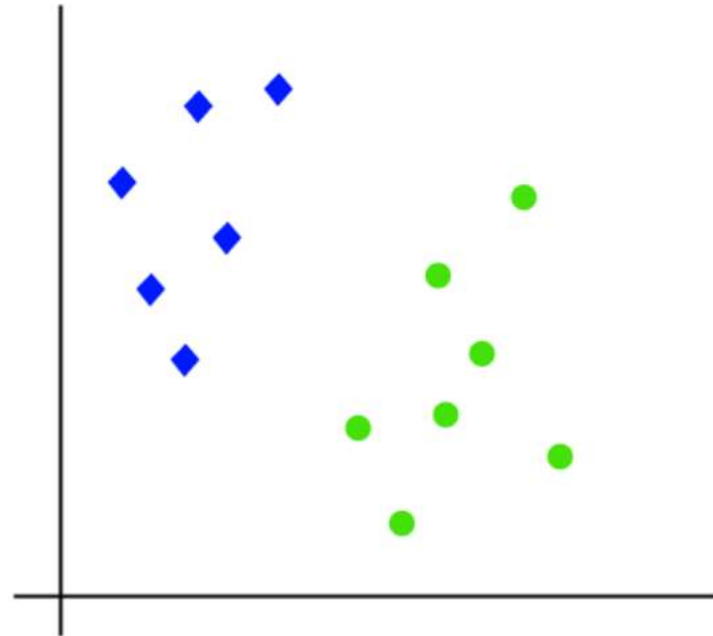
- The dimensions of the hyperplane depend on the features present in the dataset, which means if there are 2 features (as shown in image), then hyperplane will be a straight line. And if there are 3 features, then hyperplane will be a 2-dimension plane.
- We always create a hyperplane that has a maximum margin, which means the maximum distance between the data points.

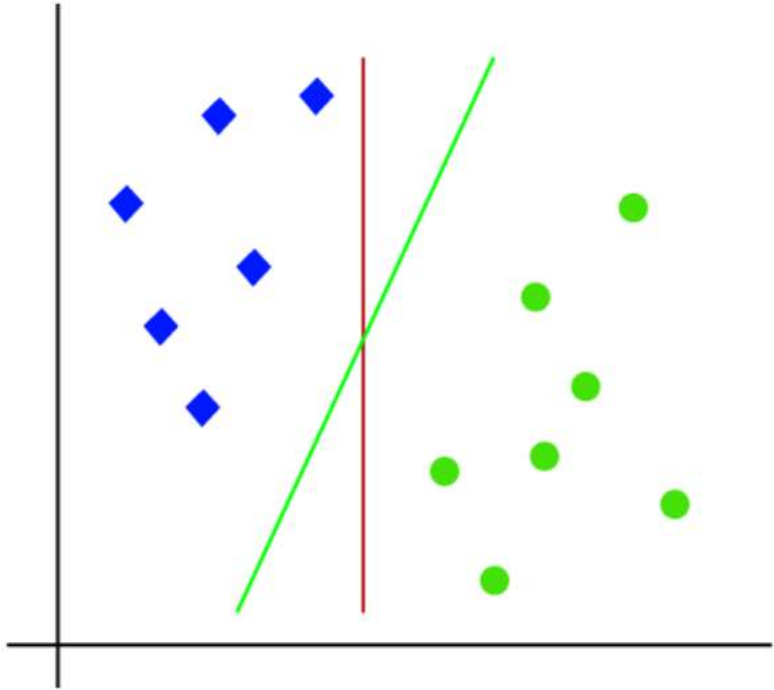
How does SVM works?

Linear SVM:

Suppose we have a dataset that has two tags (green and blue), and the dataset has two features x_1 and x_2 . We want a classifier that can classify the pair(x_1 , x_2) of coordinates in either green or blue.

Consider the below image:



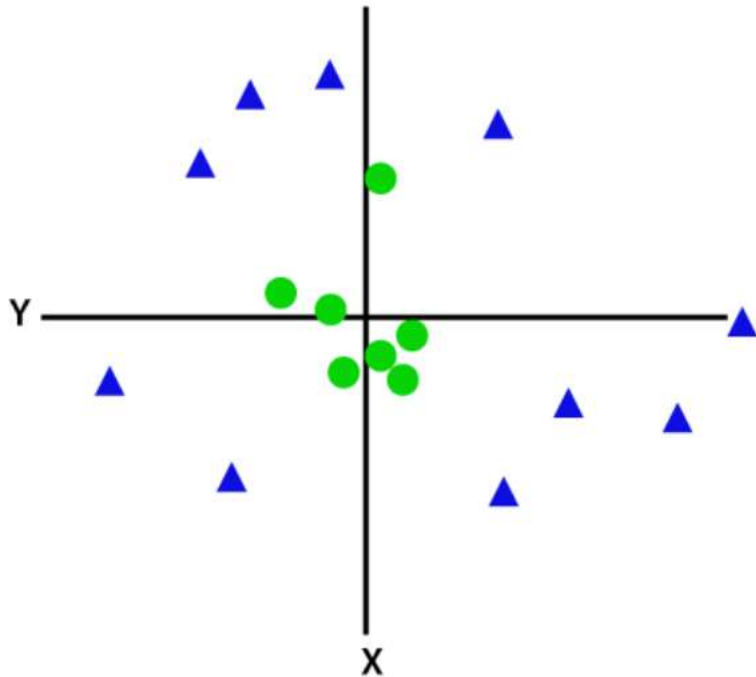


Note: There can be many hyperplanes can be drawn to separate the points

SVM algorithm helps to find the best line or decision boundary; this best boundary or region is called as a hyperplane. SVM algorithm finds the closest point of the lines from both the classes. These points are called support vectors. The distance between the vectors and the hyperplane is called as margin. And the goal of SVM is to maximize this margin. The hyperplane with maximum margin is called the optimal hyperplane.

Non-linear SVM

- If data is linearly arranged, then we can separate it by using a straight line, but for non-linear data, we cannot draw a single straight line



So to separate these data points, we need to add one more dimension. For linear data, we have used two dimensions x and y , so for non-linear data, we will add a third dimension z . It can be calculated as:

$$z = x^2 + y^2$$

